

18. On a Work by K. Hentzelt, Who Fell in the War, on Elimination Theory.

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11th Session, Friday, September 23, 1921, at 11 a.m.

(Chair: Loewy.)

1. E. Noether: On a work by K. Hentzelt, who fell in the war, on elimination theory.

Kurt Hentzelt has been missing near Dixmuiden since October 1914 and must be presumed dead. I intend shortly to publish in the *Mathematische Annalen*, in a freely reworked form, the most essential parts of his Erlangen dissertation, which he left behind complete in structure but presented almost unintelligibly.

The issue is the *general problem of elimination theory*: to determine the common zeros of all polynomials belonging to a polynomial ideal $\mathfrak{M}$. If the variables are first subjected to a linear transformation with indeterminate coefficients, the main result can be stated as follows:

A unique resultant form can be associated with the ideal $\mathfrak{M}$:

$$R_{\mathfrak{M}} = R^{(1)}(x_1 \dots x_n) \cdots R^{(i)}(x_i \dots x_n) \cdots R^{(n)}(x_n) \equiv 0(\mathfrak{M}),$$

in such a way that $R_{\mathfrak{M}}$ vanishes at all zeros of $\mathfrak{M}$ and at no others. If $\mathfrak{N}$ divides $\mathfrak{M}$ and $R_{\mathfrak{N}}$ and $R_{\mathfrak{M}}$ coincide, then $\mathfrak{N}$ and $\mathfrak{M}$ themselves coincide.

The second part of the theorem shows that the resultant form yields the zeros with their characteristic multiplicities. It is based on viewing $\mathfrak{M}$ as a module of linear forms in the power products of $x_1, \dots, x_{i-1}$; once $x_{i+1}, \dots, x_n$ have been adjoined to the coefficient domain, the individual factors $R^{(i)}(x_i \dots x_n)$ of $R_{\mathfrak{M}}$ become the norms, with respect to this module, of the corresponding “fundamental module.” Bringing in abstract ideal theory gives the following interpretation: if $\mathfrak{M} = [\mathfrak{Q}, \mathfrak{Q}_1, \dots, \mathfrak{Q}_r] = [\mathfrak{Q}, \mathfrak{L}]$ is a decomposition of $\mathfrak{M}$ into primary ideals $\mathfrak{Q}_i$, with $\mathfrak{L}$ the complement of $\mathfrak{Q}$, then to the ideal $\mathfrak{Q}$ there corresponds a primary factor $Q^{(i)}(x_i \dots x_n)$ of $R^{(i)}(x_i \dots x_n)$ which, in the sense specified above, represents the norm of $\mathfrak{L}$ with respect to $\mathfrak{Q}$.